

Jiazhen Hong

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PROFILE

AI Research Scientist @ EMOTIV | Ph.D. in Computer Engineering @ Rutgers University
Foundation Models · Distributed GPU Training · Long-Context Modeling · LLMs · Local Model Deployment
Applications: Healthcare AI · LLM-Integrated Systems · Biomedical & Time-Series Data · Assistive AI

TECHNICAL SKILLS

- **Machine Learning:** Foundation Models, LLMs, Self-Supervised Learning, Multimodal Learning, Representation Learning
- **Architectures & Frameworks:** Transformers, State Space Models (Mamba), PyTorch, TensorFlow, Hugging Face
- **Training & Systems:** Multi-GPU Training, Distributed Data Parallel (DDP), Model Optimization, GCP, Linux/SSH, Docker
- **Biomedical AI:** EEG, PPG, Biosignal Processing, Brain-Computer Interfaces, Real-Time Neural Decoding

PROFESSIONAL EXPERIENCE

EMOTIV, San Francisco, CA, USA <i>AI Research Scientist</i>	09/2026 – Present
• Develop foundation models and long-context architectures for biomedical and neural time-series data • Build and optimize distributed multi-GPU training pipelines for large-scale model pretraining • Develop self-supervised representation learning for transfer across datasets and downstream tasks • Benchmark scalable ML systems across model design, infrastructure, evaluation, and deployment	
<i>Research Intern</i>	10/2024 – 09/2025
• Developed large-scale foundation models for neural signals using distributed cloud GPU training • Published 1st-author papers at NeurIPS and IEEE ICDM workshops	
Rutgers University, Integrated Systems & Neuroimaging Lab <i>Graduate Research Assistant</i>	09/2021 – 08/2026
• Developed locally deployed LLM-integrated real-time BCI speller systems (ChatBCI) for assistive communication • Designed representation learning methods for multimodal neural and biomedical time-series data • Designed human-subject experimental paradigms, prepared eIRB/IRB protocols, and collected physiological and neural data • Led an NSF-funded research project combining machine learning and neural signal processing	
<i>Faculty Lecturer</i>	05/2022 – 07/2022
• Taught Linear Systems and Signals to a class of 30-40 students	
<i>Teaching Assistant</i>	01/2020 – 06/2025
• Digital Signal Processing / Linear Systems and Signals	
Harvard Medical School <i>Student Researcher</i>	01/2019 – 12/2020
• Performed biostatistical and machine learning analysis of protein biomarkers for early cancer detection • Applied statistical hypothesis testing and multiple-comparison correction to identify predictive biomarkers	

EDUCATION

Rutgers University, New Brunswick, NJ, USA <i>Ph.D.</i>	01/2020 – 08/2026
Computer Engineering, G.P.A.: 3.9/4.0 Thesis Topic: Artificial Intelligence for Time-Series Signal Processing in Brain-Computer Interfaces (BCIs)	
Stevens Institute of Technology, Hoboken, NJ, USA <i>M.Sc.</i>	09/2017 – 01/2019
Electrical Engineering, G.P.A.: 3.9/4.0	
Jimei & Chung Yuan Christian University, China <i>B.Sc. (Double Degree)</i>	09/2012 – 06/2016
Communication Engineering & Business, G.P.A.: 3.5/4.0 & 4.0/4.0	

ACADEMIC SERVICE & ACTIVITIES

Reviewer
• Conference: NeurIPS (2025-2026), AISTATS (2023-2024), IEEE ICASSP (2024) • Journal: Scientific Reports (2026), IEEE TBME (2025-2026), Neurocomputing (2026), Ann. N.Y. A. Sci. (2026), PLOS ONE (2026)
Technical Program Committee
• IEEE International Symposium on Biomedical Imaging (2024-2026)

PUBLICATIONS

Journal (Accepted/Published)

- **J. Hong**, W. Qian, Y. Chen, and Y. Zhang, "A geometric approach to k -means," *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, accepted, 2025
- **J. Hong**, W. Wang, and L. Najafizadeh, "ChatBCI, a P300 speller BCI with context-driven word prediction leveraging large language models, from concept to evaluation," *Scientific Reports (Nature Portfolio)*, accepted, 2025
- **J. Hong**, P. Rao, W. Wang, and L. Najafizadeh, "ChatBCI-Assist: An Intent-Based P300 Speller with A Locally-Deployed LLM and Adaptive Stopping Strategy Enabling Record Online Spelling Performance," *IEEE Transactions on Biomedical Engineering (TBME)*, accepted, 2026, **Invited Submission**

Conference (Accepted/Published)

- **J. Hong**, G. Mackellar, S. Ghane, "SAMBA: Toward an EEG Foundation Model via Differential Mamba and Coordinate-Driven Spatial Embedding," *40th Conference on Neural Information Processing Systems (NeurIPS 2026)*, *NeurReps Proceedings Track*
- **J. Hong**, and L. Najafizadeh "Improving Spelling Performance of LLM-based P300 Speller BCIs using Circular Nonadjacent Flashing and Cross-Validated Calibration," *48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2026)*, **Sole North America Finalist, Student Paper Competition**
- P. Rao, **J. Hong**, W. Wang, and L. Najafizadeh, "Llama-4-ALS: A Locally Deployable LLM, Tuned with LoRA and Dynamic Hierarchical Data Augmentation, for Speller-Based AAC," *48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2026)*
- A. Subramanian, S. Mai, **J. Hong**, and L. Najafizadeh, "Neural Dynamics of Time Perception Under Acute Pain: An Auditory Temporal Bisection Study," *23rd IEEE International Symposium on Biomedical Imaging (ISBI 2026)*
- **J. Hong**, G. Mackellar, S. Ghane, "SpellerSSL: Self-Supervised Learning with P300 Aggregation for Speller BCIs", *39th Conference on Neural Information Processing Systems (NeurIPS 2025)*, *Data on the Brain & Mind Workshop*
- **J. Hong**, G. Mackellar, S. Ghane, "An Efficient Self-Supervised Framework for Long-Sequence EEG Modeling," *IEEE International Conference on Data Mining (ICDM 2025)*, *AI for Time Series Proceedings Track*
- **J. Hong**, and L. Najafizadeh, "Enhancing Typing Speed in LLM-based P300 Speller BCIs Using A New Data-Driven Dynamic Stopping Strategy," *IEEE EMBS 12th Annual International Conference on Neural Engineering (NER 2025)*
- S. Mai, **J. Hong**, T. Shors, and L. Najafizadeh, "Longest Sustained Cortical Activity Duration in Temporal Bisection Task Aligns with Geometric Mean for Log-Spaced Probes," *IEEE EMBS 12th Annual International Conference on Neural Engineering (NER 2025)*
- **J. Hong**, P. Rao, W. Wang, S. Chen, and L. Najafizadeh, "ChatBCI4ALS: A High-Performance, LLM-Driven Intent-Based BCI Communication System for Individuals with ALS," *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2025)*, **Named Top 7% of Submissions**
- **J. Hong** and L. Najafizadeh, "TopoEEG: a TimeSformer-Based Topographic Image Representation Method for Early Single-Trial Detection of P300," *22nd IEEE International Symposium on Biomedical Imaging (ISBI 2025)*
- **J. Hong**, L. Najafizadeh, "P3T: A Transformer Model for Enhancing Character Recognition Rates in P300 Speller Systems," *58th Annual Asilomar Conference on Signals, Systems, and Computers (Asilomar 2024)*
- **J. Hong**, F. Shamsi, and L. Najafizadeh, "A Deep Learning Framework Based on Dynamic Channel Selection for Early Classification of Left and Right Hand Motor Imagery Tasks," *44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2022)*

(Under review)

- **J. Hong**, S. Mai, and L. Najafizadeh, "CORA: Coordinate-Aware ROI Attention for EEG Representation Learning in Cortical Space," (Journal)
- **J. Hong**, W. Wang, S. Haghani, and L. Najafizadeh, "Subject-specific Channel Selection Based on Davies-Bouldin Index for EEG Motor Imagery Classification," (Journal)
- **J. Hong**, and L. Najafizadeh, "A Geometric Spatio-Learnable EEG Foundation Model Across Visual and Auditory Modalities in the Temporal Bisection Task" (Journal)

AWARDS

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| - Sole North America Finalist, EMBC 2026 Student Paper Competition | 07/2026 |
| - ECE Development and Research Excellence Award (2025-2026 cycle), Rutgers University | 01/2026 |
| - Research & Conference Travel Award, funded by the School of Graduate Studies (SGS), Rutgers University | 03/2025 |
| - Trainee Travel Award of 2024-2025 Rutgers Brain Health Institute, Cognitive and Sensory Neuroscience Focus Area | 02/2025 |
| - Travel Award of 2024 IEEE Brain Discovery & Neurotechnology Workshop | 09/2024 |
| - ECE Best Teaching Assistant Award for Fall 2023, Rutgers University | 05/2024 |
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